

Objective

The objective of this task is to develop a **Palindrome Checker** using Python and Streamlit. The application accepts a word or sentence from the user and determines whether it remains the same when read forward and backward. The task helps develop practical knowledge of string manipulation, string slicing, comparison logic, input validation, and text normalization.

Key Features

- Accepts a word or complete sentence.
- Converts input to lowercase.
- Ignores unnecessary spaces.
- Reverses the normalized string using Python slicing.
- Compares the original normalized text with its reversed version.
- Displays whether the input is a palindrome.
- Handles empty input.
- Displays the normalized and reversed text.
- Provides example palindrome and non-palindrome values.
- Uses an interactive Streamlit interface.

Technologies Used

The project is developed using **Python** and **Streamlit**. Python string methods such as `lower()` and `split()` are used for normalization, while string slicing `[::-1]` is used to reverse the input. Conditional statements are used to compare the normalized and reversed strings and produce the appropriate result.

Learning Outcomes

This task provides practical experience with Python strings, string slicing, comparison operations, conditional statements, user input handling, and basic text normalization. It also improves understanding of how simple Python logic can be integrated into an interactive Streamlit application.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY12>

```
app.py > ...
1  import streamlit as st
2
3  # Page configuration
4  st.set_page_config(
5      page_title="Palindrome Checker",
6      page_icon="🔍",
7      layout="centered"
8  )
9
10 # Custom CSS
11 st.markdown("""
12 <style>
13     .title {
14         text-align: center;
15         font-size: 40px;
16         font-weight: bold;
17     }
18
19     .subtitle {
20         text-align: center;
21         font-size: 18px;
22         margin-bottom: 25px;
23     }
24
25     .result {
26         padding: 20px;
27         border-radius: 10px;
28         text-align: center;
29         font-size: 24px;
30         font-weight: bold;
31     }
32 </style>
33 """, unsafe_allow_html=True)
34
35 # Header
36 st.markdown(
37     '<div class="title">🔍 Palindrome Checker</div>',
38     unsafe_allow_html=True
39 )
40
41 st.markdown(
42     '<div class="subtitle">'
43     'Check whether a word or sentence reads the same forward and backward'
44     '</div>',
45     unsafe_allow_html=True
46 )
47
48 st.divider()
49
50 # Input
51 st.subheader("🔍 Enter a Word or Sentence")
52
53 text = st.text_input(
54     "Enter your text:",
55     placeholder="Example: Madam or Never odd or even"
56 )
57
58 # Check button
59 if st.button("🔍 Check Palindrome", use_container_width=True):
60
61     if not text.strip():
62         st.warning("⚠️ Please enter a word or sentence.")
63     else:
64         # Normalize input:
65         # Convert to lowercase and remove spaces
66         normalized_text = "".join(text.lower().split())
67
68         # Reverse the normalized string
69         reversed_text = normalized_text[::-1]
70
71         # Compare original and reversed strings
72         if normalized_text == reversed_text:
73             st.success("🎉 Yes! It is a palindrome.")
74
75             st.markdown(
76                 '<div class="result">✅ PALINDROME</div>',
77                 unsafe_allow_html=True
78             )
79         else:
80             st.error("❌ No! It is not a palindrome.")
81
82             st.markdown(
83                 '<div class="result">❌ NOT A PALINDROME</div>',
84                 unsafe_allow_html=True
85             )
86
87     st.divider()
88
89 # Show processing details
90 st.subheader("📊 Analysis")
91
92 col1, col2 = st.columns(2)
```

<http://localhost:8501>

🔍 Palindrome Checker

Check whether a word or sentence reads the same forward and backward

🔍 Enter a Word or Sentence

Enter your text:

malayalam

🔍 Check Palindrome

🎉 Yes! It is a palindrome.

✅ PALINDROME

📊 Analysis ∞

Original Text:

malayalam

Normalized Text:

malayalam

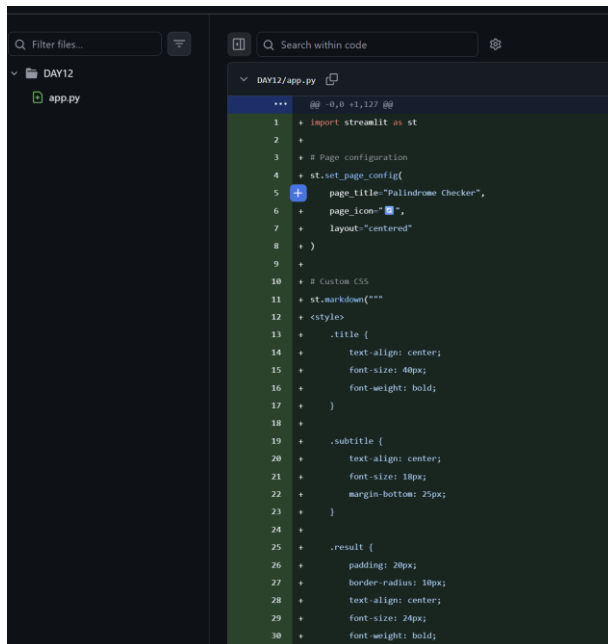
Reversed Text:

malayalam

💡 Examples

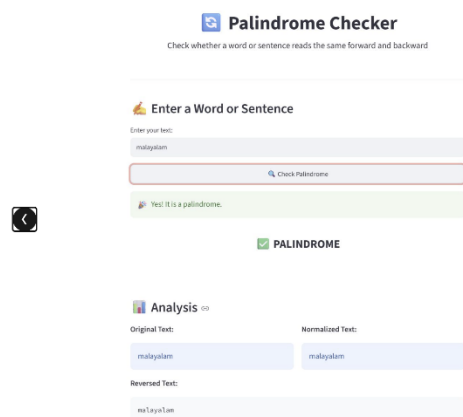
- ✅ Madam → Palindrome
- ✅ Racecar → Palindrome
- ✅ Never odd or even → Palindrome
- ❌ Python → Not a palindrome

<https://github.com/shambhavi0804/VEDA-internship/commit/a98d26e9d551c57f48d5a8d8c579e82565c6b487>



```
... @ -0.0 -1,127 @@
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```

<https://lnkd.in/p/dTE25Acm>



Palindrome Checker
Check whether a word or sentence reads the same forward and backward

Enter a Word or Sentence

Enter your text:
malayalam

Check Palindrome

Yes! It is a palindrome.

✅ PALINDROME

Analysis

Original Text: malayalam

Normalized Text: malayalam

Reversed Text: malayalam

